

- 8 (a) State the name of the class (group) of fundamental particles that contains a neutrino.

..... [1]

- (b) A hadron P has a charge of $+1e$, where e is the elementary charge. The hadron P is composed of a down antiquark and only one other quark.

- (i) Identify a possible flavour for this other quark.

..... [1]

- (ii) State what type of hadron is P.

..... [1]

- (c) Nucleus Q undergoes radioactive decay to form nucleus R, emitting an antineutrino and another particle X, as shown in the decay equation.



- (i) State what particle is represented by X.

..... [1]

- (ii) Compare the nucleon numbers of Q and R.

..... [1]

- (iii) Compare the charges of Q and R.

..... [1]

[Total: 6]

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